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BSCS - C301

Finals Task 03. DAX TIME INTELLIGENCE

Overview:

In this lab, you'll create measures with DAX expressions that involve time intelligence.

In this lab, learn how to:

- Use various time intelligence functions to manipulate filter context that specific concerns dates.

This lab should take approximately 15 minutes.

1. SALES YTD CALCULATION

The screenshot shows the Power BI Desktop interface. On the left, the DAX formula bar contains the following code:

```
1 Sales YTD =
2 TOTALYTD(
3     SUM(Sales[Sales]),
4     'Date'[Date],
5     "y-y")
6
7
```

On the right, a table visualization displays the results of the measure. The table has columns for Year, Month, Sales, and Sales YTD. The data is filtered by Year (2020) and Month (Oct, Nov, Dec). The Sales YTD column shows the cumulative sales for each month and the total for the year.

Year	Month	Sales	Sales YTD
2020	Oct	\$748.68	\$748.68
2020	Nov	\$655.59	\$1,404.27
2020	Dec	\$758.93	\$2,163.20
2021	Jan	\$741.85	\$2,905.05
2021	Feb	\$677.45	\$3,582.50
2021	Mar	\$752.31	\$4,334.81
2021	Apr	\$734.58	\$5,069.39
2021	May	\$808.94	\$5,878.33
2021	Jun	\$552.95	\$6,431.28
2021	Jul	\$391.81	\$6,823.09
2021	Aug	\$372.75	\$7,195.84
2021	Sep	\$365.30	\$7,561.14
2021	Oct	\$382.00	\$7,943.14
2021	Nov	\$412.48	\$8,355.62
2021	Dec	\$407.08	\$8,762.70
Total		\$4,466.39	\$8,762.70

2. SALES YoY Growth (Percentage Logic)

The screenshot shows the Power BI Desktop interface. On the left, the DAX formula bar contains the following code:

```
1 Sales YoY Growth =
2 VAR SalesPriorYear =
3     CALCULATE(
4         SUM(Sales[Sales]),
5         PARALLELPERIOD(
6             'Date'[Date],
7             -12,
8             MONTH
9         )
10    )
11 RETURN
12     DIVIDE(
13         (SUM(Sales[Sales]) - SalesPriorYear),
14         SalesPriorYear
15    )
16
```

On the right, a table visualization displays the results of the measure. The table has columns for Year, Month, Sales, and Sales YoY Growth. The data is filtered by Year (2017) and Month (Jul, Aug, Sep, Oct). The Sales YoY Growth column shows the percentage growth for each month and the total for the year.

Year	Month	Sales	Sales YoY Growth
2017	Jul	\$655.59	252%
2017	Aug	\$758.93	785%
2017	Sep	\$741.85	593%
2017	Oct	\$677.45	499%

3. Final Output

Year	Avg Price	Median Price	Min Price	Max Price	Orders	Order Lines	Sum of Sales	Sales YTD	Sales YoY Growth
FY2021									-100.00%
2020 Jul									-100.00%
2020 Aug									-100.00%
2020 Sep									-100.00%
2020 Oct									-100.00%
2020 Nov									-100.00%
2020 Dec									-100.00%
2021 Jan									-100.00%
2021 Feb									-100.00%
2021 Mar									-100.00%
2021 Apr									-100.00%
2021 May									-100.00%
FY2018	\$748.68	\$419.46	\$4.75	\$2,146.96	739	8,459	\$16,429,043	₱16,429,042.6	
2017 Jul	\$655.59	\$419.46	\$5.19	\$2,146.96	38	352	\$489,328	₱489,328.4	
2017 Aug	\$758.93	\$419.46	\$4.75	\$2,146.96	75	785	\$1,540,072	₱2,029,400.42	
2017 Sep	\$741.85	\$419.46	\$5.19	\$2,146.96	60	593	\$1,166,332	₱3,195,732.87	
2017 Oct	\$677.45	\$419.46	\$5.19	\$2,146.96	40	499	\$844,833	₱4,040,565.58	
2017 Nov	\$752.31	\$419.46	\$5.01	\$2,146.96	90	1,106	\$2,325,755	₱6,366,320.49	
2017 Dec	\$734.58	\$419.46	\$5.01	\$2,146.96	63	803	\$1,703,435	₱8,069,755.58	
2018 Jan	\$808.94	\$419.46	\$5.19	\$2,146.96	40	377	\$713,230	₱8,782,985.12	
2018 Feb	\$896.80	\$419.46	\$5.01	\$2,146.96	79	866	\$1,900,794	₱10,683,779.58	
2018 Mar	\$863.54	\$419.46	\$5.19	\$2,146.96	64	653	\$1,455,280	₱12,139,059.85	
2018 Apr	\$732.25	\$419.46	\$5.19	\$2,146.96	37	494	\$883,011	₱13,022,070.83	
2018 May	\$761.30	\$419.46	\$4.75	\$2,146.96	85	1,112	\$2,269,720	₱15,291,790.7	
2018 Jun	\$552.95	\$419.46	\$4.75	\$2,146.96	68	819	\$1,137,252	₱16,429,042.6	
FY2019	\$397.81	\$202.33	\$4.32	\$1,466.01	1,255	21,670	\$27,979,780	₱27,979,779.53	70.31%
2018 Jul	\$350.74	\$196.33	\$4.75	\$1,466.01	72	1,723	\$2,411,559	₱2,411,558.85	392.83%
2018 Aug	\$372.75	\$202.33	\$4.75	\$1,466.01	139	2,964	\$3,615,914	₱6,027,473.31	134.79%
2018 Sep	\$365.30	\$196.33	\$4.32	\$1,466.01	111	2,185	\$2,894,647	₱8,922,119.92	148.18%
2018 Oct	\$382.00	\$202.33	\$5.19	\$1,466.01	73	1,406	\$1,804,177	₱10,726,296.85	113.55%
2018 Nov	\$412.48	\$209.26	\$5.01	\$1,466.01	133	2,345	\$3,054,997	₱13,781,293.57	31.36%
2018 Dec	\$407.08	\$202.33	\$4.75	\$1,466.01	114	1,732	\$2,188,206	₱15,969,499.23	28.46%
Total	\$446.39	\$214.24	\$1.33	\$2,146.96	3,616	57,851	\$77,548,570		0.00%